

Strategic Risk & Severity Analysis Report

A Data-Driven Evaluation Using Advanced SQL Modeling

Executive Summary

This report evaluates national accident patterns to identify structural risk trends, geographic concentration, and high-severity drivers. The objective is to move beyond raw accident counts and uncover underlying severity dynamics, risk escalation patterns, and fatality exposure.

Key findings indicate measurable year-over-year fluctuation, concentrated risk contribution from specific states, and elevated severity under certain time and weather conditions. A rolling severity model further reveals whether risk is structurally increasing rather than temporarily fluctuating.

The analysis provides a scalable framework for targeted policy intervention, resource allocation, and high-impact risk mitigation.

Business Problem Statement

Despite ongoing road safety efforts, accident frequency and severity remain persistent concerns. Decision-makers require clarity on:

- Whether accident volumes are structurally increasing or stabilizing
- Which regions contribute disproportionately to total risk
- Where fatal exposure is highest
- Which environmental and time-based factors intensify severity

Without structured analytical modeling, responses remain reactive rather than strategic.

This project addresses these gaps using trend modeling, severity scoring, concentration analysis, and multi-dimensional risk evaluation.

Data Overview

The dataset contains multi-year road accident records with geographic, temporal, and environmental attributes.

Key variables analyzed include:

- Year
- State_Name
- Accident_Severity
- Time_of_Day
- Weather_Conditions

Feature engineering was applied to standardize severity scoring and segment time into operational risk buckets, enabling structured modeling.

Analytical Framework

YoY Growth Rate

Finding: Recent year shows a measurable decline in accident volume, indicating possible policy impact.

Impact: The decline in volume suggests an improvement in preventive measures and road safety strategies, although further analysis of incident severity is required to determine the overall effectiveness of current policies.

Recommendation: Focus on identifying the specific interventions that led to the 2023 reduction and strengthen these successful protocols to maintain the downward trend in future periods.

	Year	total_accidents	yoy_change	yoy_growth_pct
▶	2018	242	NULL	NULL
	2019	237	-5	-2.07
	2020	270	33	13.92
	2021	268	-2	-0.74
	2022	296	28	10.45
	2023	252	-44	-14.86

Severity Trend Insight

Finding:

The 3-year rolling data indicates a notable **reduction** in accident volume in the most recent period, reversing the previous year's spike and suggesting a cooling trend in incident frequency.

Impact:

A decline in volume suggests that current preventive measures may be gaining traction; however, the lack of a steady downward trend over the long term indicates that safety remains sensitive to external factors.

Recommendation:

Shift focus toward analyzing the **severity** of the remaining 252 incidents. If the volume is dropping but the severity remains high, resources should be pivoted from "broad prevention" to "fatality reduction" and specialized emergency response in high-risk zones.

	Year	rolling_3yr_severity
▶	2018	2.06
	2019	2.03
	2020	2.01
	2021	1.99
	2022	1.99
	2023	2.01

Geographic Risk Concentration

Finding:Accident cases are unevenly distributed across states, with certain states such as Karnataka, Puducherry, Goa, and Haryana showing relatively higher contributions compared to others.

Impact:Road accident risk is geographically concentrated, indicating that some regions face higher safety challenges than the national average.

	State_Name	total_accidents	cumulative_pct
▶	Uttarakhand	59	3.77
	Karnataka	57	7.41
	Puducherry	56	10.99
	Goa	55	18.02
	Haryana	55	18.02
	Maharashtra	54	28.37
	Nagaland	54	28.37
	Sikkim	54	28.37
	Delhi	53	35.14
	Jammu and Kashmir	53	35.14
	Kerala	52	38.47
	Odisha	51	44.98
	Rajasthan	51	44.98

Recommendation:Prioritize enforcement efforts, infrastructure improvements, and road safety awareness programs in the higher-contributing states to achieve the greatest reduction in overall accidents

Fatality Risk Assessment

	State_Name	fatal_count	total_accidents	fatal_ratio_pct
▶	Bihar	19	43	44.19
	West Bengal	21	49	42.86
	Maharashtra	23	54	42.59
	Nagaland	23	54	42.59
	Tamil Nadu	19	46	41.30
	Assam	18	46	39.13
	Chhattisgarh	18	46	39.13
	Gujarat	19	49	38.78
	Goa	21	55	38.18
	Uttarakhand	22	59	37.29
	Punjab	17	46	36.96
	Andhra Pra...	15	41	36.59
	Kerala	19	52	36.54

Result 6 ×

Finding: Several states (e.g., Bihar, West Bengal, Maharashtra, and Nagaland) exhibit relatively high fatality ratios despite having moderate total accident counts.

Impact: A high fatality ratio suggests deeper road safety concerns — such as delayed emergency response, unsafe road design, poor helmet/seatbelt compliance, or high-speed driving conditions — leading to more severe accident outcomes.

Recommendation: Prioritize fatality reduction strategies in high fatal-ratio states by strengthening trauma care access, enforcing safety gear compliance, and conducting targeted road infrastructure and speed management audits.

High-Risk Condition Modeling

Finding: Accident risk scores peak under specific combinations of state, time of day, and weather conditions, indicating identifiable high-risk scenarios rather than uniformly distributed risk.

State_Name	time_bucket	Weather_Conditions	accidents	risk_score
Chhattisgarh	Morning	Foggy	4	3.00
Gujarat	Afternoon	Stormy	1	3.00
Odisha	Evening	Hazy	1	3.00
Assam	Night	Stormy	4	3.00
Chhattisgarh	Evening	Clear	2	3.00
Chhattisgarh	Morning	Clear	1	3.00
Maharashtra	Evening	Clear	1	3.00
Haryana	Evening	Clear	2	3.00
Karnataka	Afternoon	Hazy	2	3.00
West Bengal	Afternoon	Foggy	1	3.00
Tamil Nadu	Morning	Clear	2	3.00
Bihar	Afternoon	Rainy	2	3.00
Maharashtra	Afternoon	Foggy	3	3.00

Impact:Accident severity is condition-dependent. Factors such as low visibility (foggy/hazy), adverse weather (stormy/rainy), and certain time periods increase the likelihood of more serious incidents.

Recommendation:Implement targeted traffic enforcement, dynamic speed controls, and focused public awareness campaigns during high-risk time-weather windows—especially in states where these patterns repeatedly appear—to proactively reduce severe accidents.